

Calculus 2 (20152-2)

Exercise Sheet 9 — Full Solutions

Line Integrals (Types I & II), Green's Theorem, Conservative Fields

Problem 1 Arc Length of the Astroid $x^{2/3} + y^{2/3} = 1$

The curve is symmetric about both axes, so it suffices to compute the arc length in the first quadrant and multiply by 4.

Parametrisation (first quadrant):

$$x(t) = \cos^3 t, \quad y(t) = \sin^3 t, \quad t \in \left[0, \frac{\pi}{2}\right].$$

Verify: $x^{2/3} + y^{2/3} = \cos^2 t + \sin^2 t = 1$. ✓

$$\dot{x} = -3\cos^2 t \sin t, \quad \dot{y} = 3\sin^2 t \cos t.$$

$$ds = \sqrt{\dot{x}^2 + \dot{y}^2} dt = \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt = 3\sqrt{\cos^2 t \sin^2 t} dt = 3\cos t \sin t dt.$$

Arc length in the first quadrant:

$$L_1 = \int_0^{\pi/2} 3\cos t \sin t dt = 3\left[\frac{\sin^2 t}{2}\right]_0^{\pi/2} = \frac{3}{2}.$$

$$\boxed{L = 4 \cdot \frac{3}{2} = 6.}$$

Problem 2 Type-I Line Integrals

(a) $\int_C xy^4 ds$, C : right half of $x^2 + y^2 = 16$

Parametrise the right semicircle ($x \geq 0$):

$$x = 4\cos t, \quad y = 4\sin t, \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad ds = 4 dt.$$

$$\int_C xy^4 ds = \int_{-\pi/2}^{\pi/2} (4\cos t)(4\sin t)^4 \cdot 4 dt = 4^6 \int_{-\pi/2}^{\pi/2} \cos t \sin^4 t dt.$$

The integrand is even, so

$$= 2 \cdot 4096 \int_0^{\pi/2} \cos t \sin^4 t \, dt = 8192 \left[\frac{\sin^5 t}{5} \right]_0^{\pi/2} = \frac{8192}{5} = \boxed{1638.4}$$

(b) $\int_C xyz \, ds, \quad C: x = 2t, y = 3 \sin t, z = 3 \cos t, 0 \leq t \leq \frac{\pi}{2}$

$$\dot{x} = 2, \quad \dot{y} = 3 \cos t, \quad \dot{z} = -3 \sin t, \quad ds = \sqrt{4 + 9 \cos^2 t + 9 \sin^2 t} \, dt = \sqrt{13} \, dt.$$

$$\int_C xyz \, ds = \sqrt{13} \int_0^{\pi/2} (2t)(3 \sin t)(3 \cos t) \, dt = 18\sqrt{13} \int_0^{\pi/2} t \sin t \cos t \, dt = 9\sqrt{13} \int_0^{\pi/2} t \sin 2t \, dt.$$

Integrate by parts ($u = t, dv = \sin 2t \, dt, v = -\frac{\cos 2t}{2}$):

$$= 9\sqrt{13} \left(\left[-\frac{t \cos 2t}{2} \right]_0^{\pi/2} + \int_0^{\pi/2} \frac{\cos 2t}{2} \, dt \right) = 9\sqrt{13} \left(\frac{\pi}{4} + \left[\frac{\sin 2t}{4} \right]_0^{\pi/2} \right) = 9\sqrt{13} \cdot \frac{\pi}{4}.$$

$$\boxed{\int_C xyz \, ds = \frac{9\pi\sqrt{13}}{4}}$$

(c) $\int_C xy \, ds, \quad C: \text{straight segment from } (-1, 1) \text{ to } (2, 3)$

Parametrise: $x = -1 + 3t, y = 1 + 2t, t \in [0, 1]; \quad ds = \sqrt{9 + 4} \, dt = \sqrt{13} \, dt.$

$$\begin{aligned} \int_C xy \, ds &= \sqrt{13} \int_0^1 (-1 + 3t)(1 + 2t) \, dt = \sqrt{13} \int_0^1 (-1 + t + 6t^2) \, dt \\ &= \sqrt{13} \left[-t + \frac{t^2}{2} + 2t^3 \right]_0^1 = \sqrt{13} \left(-1 + \frac{1}{2} + 2 \right) = \boxed{\frac{3\sqrt{13}}{2}} \end{aligned}$$

(d) $\int_C \frac{y}{\sqrt{x}} \, ds, \quad C: \text{arc of } y^2 = \frac{4}{9}x^3 \text{ from } A(3, 2\sqrt{3}) \text{ to } B(8, \frac{32\sqrt{2}}{3})$

Taking the positive branch: $y = \frac{2}{3}x^{3/2}.$

$$\frac{y}{\sqrt{x}} = \frac{\frac{2}{3}x^{3/2}}{\sqrt{x}} = \frac{2}{3}x, \quad y' = x^{1/2}, \quad ds = \sqrt{1 + x} \, dx.$$

$$\int_C \frac{y}{\sqrt{x}} \, ds = \int_3^8 \frac{2}{3}x\sqrt{1+x} \, dx.$$

Substitute $u = 1 + x$, limits $4 \rightarrow 9$:

$$\begin{aligned} &= \frac{2}{3} \int_4^9 (u-1)\sqrt{u} \, du = \frac{2}{3} \int_4^9 (u^{3/2} - u^{1/2}) \, du = \frac{2}{3} \left[\frac{2u^{5/2}}{5} - \frac{2u^{3/2}}{3} \right]_4^9 \\ &= \frac{2}{3} \left[\left(\frac{486}{5} - 18 \right) - \left(\frac{64}{5} - \frac{16}{3} \right) \right] = \frac{2}{3} \cdot \frac{1076}{15} = \boxed{\frac{2152}{45}} \end{aligned}$$

Problem 3 $\int_C \vec{F} \cdot d\vec{r}$

$$\vec{F}(x, y) = e^x \hat{i} + xy \hat{j}, \quad \vec{r}(t) = t^2 \hat{i} + t^3 \hat{j}, \quad 0 \leq t \leq 1.$$

$$\vec{r}'(t) = 2t \hat{i} + 3t^2 \hat{j}, \quad \vec{F}(\vec{r}(t)) = e^{t^2} \hat{i} + t^5 \hat{j}.$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (2te^{t^2} + 3t^7) dt = \left[e^{t^2} \right]_0^1 + \left[\frac{3t^8}{8} \right]_0^1 = (e - 1) + \frac{3}{8} = \boxed{\frac{8e - 5}{8}}$$

Problem 4 Type-II Line Integrals

(a) $\int_C \sin x \, dx$, C : arc of $x = y^4$ from $A(1, -1)$ to $B(1, 1)$

Since $P \, dx + Q \, dy = \sin x \, dx = d(-\cos x)$ is an exact differential, the integral depends only on the endpoints' x -coordinates. Both A and B have $x = 1$, so:

$$\int_C \sin x \, dx = [-\cos x]_{x=1}^{x=1} = \boxed{0}.$$

Alternative check: Parametrise by $y \in [-1, 1]$: $dx = 4y^3 \, dy$ and $\sin(y^4) \cdot 4y^3$ is an odd function of y , giving zero over $[-1, 1]$.

(b) $\int_C xy \, dx + (x - y) \, dy$, C : $O(0, 0) \rightarrow A(2, 0) \rightarrow B(3, 2)$

Segment OA ($y = 0$, $x : 0 \rightarrow 2$, $dy = 0$):

$$\int_{OA} = \int_0^2 x \cdot 0 \, dx = 0.$$

Segment AB : parametrise $x = 2 + t$, $y = 2t$, $t \in [0, 1]$; $dx = dt$, $dy = 2 \, dt$.

$$\begin{aligned} \int_{AB} &= \int_0^1 [(2+t)(2t) \cdot 1 + (2+t-2t) \cdot 2] dt = \int_0^1 [4t + 2t^2 + 4 - 2t] dt \\ &= \int_0^1 [2t + 2t^2 + 4] dt = \left[t^2 + \frac{2t^3}{3} + 4t \right]_0^1 = 1 + \frac{2}{3} + 4 = \boxed{\frac{17}{3}} \end{aligned}$$

Problem 5 Closed-Curve Type-II Integrals (Green's Theorem)

(a) $\oint_C y^2 dx + x^2 dy$, C : triangle $x = 0$, $y = 0$, $x + y = 1$, CCW

Green's theorem with $P = y^2$, $Q = x^2$:

$$\begin{aligned}\oint_C &= \iint_D (2x - 2y) dA = 2 \int_0^1 \int_0^{1-x} (x - y) dy dx. \\ 2 \int_0^1 \left[xy - \frac{y^2}{2} \right]_0^{1-x} dx &= 2 \int_0^1 \left[x(1-x) - \frac{(1-x)^2}{2} \right] dx = \int_0^1 (1-x)(3x-1) dx \\ &= \int_0^1 (-3x^2 + 4x - 1) dx = \left[-x^3 + 2x^2 - x \right]_0^1 = \boxed{0}\end{aligned}$$

(b) $\oint_C x^2 y dx + xy^3 dy$, C : unit square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, CCW

Green's theorem with $P = x^2 y$, $Q = xy^3$:

$$\begin{aligned}\oint_C &= \iint_{[0,1]^2} (y^3 - x^2) dA = \int_0^1 \int_0^1 (y^3 - x^2) dy dx = \int_0^1 \left[\frac{y^4}{4} - x^2 y \right]_0^1 dx = \int_0^1 \left(\frac{1}{4} - x^2 \right) dx. \\ &= \left[\frac{x}{4} - \frac{x^3}{3} \right]_0^1 = \frac{1}{4} - \frac{1}{3} = \boxed{-\frac{1}{12}}\end{aligned}$$

(c) $\oint_C (x^2 + y^2) dx + 2xy dy$, C : arc of $y = x^2$ from $(0, 0)$ to $(2, 4)$, then $(2, 4) \rightarrow (0, 4) \rightarrow (0, 0)$

With $P = x^2 + y^2$ and $Q = 2xy$:

$$\frac{\partial Q}{\partial x} = 2y = \frac{\partial P}{\partial y}.$$

The field is conservative (exact), so every closed-curve integral is zero:

$$\oint_C (x^2 + y^2) dx + 2xy dy = \boxed{0}.$$

(d) $\oint_C xy^3 dx + x^2 y^2 dy$, C : boundary of region bounded by $y = x^2$ and $y = \sqrt{x}$, CCW

Green's theorem with $P = xy^3$, $Q = x^2 y^2$:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2xy^2 - 3xy^2 = -xy^2.$$

Region D : $0 \leq x \leq 1$, $x^2 \leq y \leq \sqrt{x}$.

$$\begin{aligned}\oint_C &= - \iint_D xy^2 dA = - \int_0^1 x \int_{x^2}^{\sqrt{x}} y^2 dy dx = - \int_0^1 x \cdot \frac{1}{3} (x^{3/2} - x^6) dx \\ &= -\frac{1}{3} \int_0^1 (x^{5/2} - x^7) dx = -\frac{1}{3} \left[\frac{2x^{7/2}}{7} - \frac{x^8}{8} \right]_0^1 = -\frac{1}{3} \left(\frac{2}{7} - \frac{1}{8} \right) = -\frac{1}{3} \cdot \frac{9}{56} = \boxed{-\frac{3}{56}}\end{aligned}$$

Problem 6 Green's Theorem Applied Directly

Compute $\oint_C (y + e^{\sqrt{x}}) dx + (2x + \cos(y^2)) dy$, where C is the boundary (CCW) of the region between $y = x^2$ and $x = y^2$.

Let $P = y + e^{\sqrt{x}}$ and $Q = 2x + \cos(y^2)$. Then:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2 - 1 = 1.$$

By Green's theorem, the integral equals the area of D :

$$\oint_C = \iint_D 1 dA = \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2x^{3/2}}{3} - \frac{x^3}{3} \right]_0^1 = \frac{2}{3} - \frac{1}{3} = \boxed{\frac{1}{3}}$$

Problem 7 Conservative Fields and Potential Functions

(a) $\vec{F}(x, y) = e^{2y} \hat{i} + (1 + 2xe^{2y}) \hat{j}$

Check conservativeness:

$$\frac{\partial P}{\partial y} = 2e^{2y} = \frac{\partial Q}{\partial x}. \quad \checkmark$$

Find potential f such that $\nabla f = \vec{F}$:

$$f_x = e^{2y} \implies f = xe^{2y} + g(y).$$

$$f_y = 2xe^{2y} + g'(y) = 1 + 2xe^{2y} \implies g'(y) = 1 \implies g(y) = y.$$

$$f(x, y) = xe^{2y} + y.$$

Evaluate: path $\vec{r}(t) = te^t \hat{i} + (1+t)\hat{j}$, $0 \leq t \leq 1$:

$$\vec{r}(0) = (0, 1), \quad \vec{r}(1) = (e, 2).$$

$$\int_C \vec{F} \cdot d\vec{r} = f(e, 2) - f(0, 1) = (e \cdot e^4 + 2) - (0 + 1) = \boxed{e^5 + 1}$$

(b) $\vec{F}(x, y) = (1 + 4x^3y^3)\hat{i} + 3x^4y^2\hat{j}$

Check conservativeness:

$$\frac{\partial P}{\partial y} = 12x^3y^2 = \frac{\partial Q}{\partial x}. \quad \checkmark$$

Find potential f :

$$f_x = 1 + 4x^3y^3 \implies f = x + x^4y^3 + g(y).$$

$$f_y = 3x^4y^2 + g'(y) = 3x^4y^2 \implies g'(y) = 0 \implies g = \text{const.}$$

$$f(x, y) = x + x^4y^3.$$

Evaluate: path $\vec{r}(t) = te^t\hat{i} + (1+t)\hat{j}$, $0 \leq t \leq 1$:

$$\vec{r}(0) = (0, 1), \quad \vec{r}(1) = (e, 2).$$

$$\int_C \vec{F} \cdot d\vec{r} = f(e, 2) - f(0, 1) = (e + e^4 \cdot 8) - 0 = \boxed{e + 8e^4}$$